

ROSHAN RAVI

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Objective

To architect and develop next-generation applications for the cloud

Work Experience

Innovate Loop

MongoDB Consultant, Major Health Insurance Firm (January 2021 – Present)

- Developed Python/MongoDB ETL platform to process daily incremental load for 300,000 members
- Optimized database infrastructure to handle comprehensive claims data for 4,000,000 members
- Leveraged multi-node Python cluster to efficiently parse large XML files into JSON and store in MongoDB within service-level agreement (SLA) of 3 hours
- Designed ETL platform architecture and presented to key stakeholders in committee review

Founder & CEO (September 2019 – Present)

- Corresponded with clients to ensure successful project deliverables
- Built long-term relationships with vendors

Nuance Communications

Software Engineer (June 2018 – April 2021)

- Worked on Dragon Medical Virtual Assistant (DMVA), using natural language understanding (NLU) to help streamline clinical workflows via conversational AI
- Architected Azure Kubernetes Service (AKS) cloud deployments
- Developed web/iOS/Android SDKs and sample applications for our REST & WebSocket APIs
- Redesigned internal CLI tools to simplify the NLU model development process
- Coordinated the integration of our solution within partner applications

Software Engineer Intern (Summer 2017)

- Developed speaker diarization technology to effectively capture doctor-patient conversations
- Built initial prototype of the company's flagship Dragon Ambient Experience (DAX) offering

Software Engineer Intern (Summer 2016)

- Created an application for Nuance's diagnostic division to assist doctors in writing PowerScribe Clinical Guidance Rules to aid creating radiology reports

The Green Alliance (FIRST Robotics Competition Match Analysis)

Founder & Lead Developer (September 2016 – June 2019)

- Led development team to implement a match analysis platform within the six-week build season
- Coordinated dozens of teams internationally to collect data about 500+ robots over 6000+ matches
- Architected custom Node.js/MongoDB platform to support decentralized data ingestion
- Leveraged MongoDB aggregation queries to develop match strategy dashboard

Education

University of Massachusetts Lowell – B.S. Computer Science

Fall 2019 – Fall 2021: Data Science Concentration, Math Minor, Commonwealth Honors Scholar

Relevant Coursework: Computing for Health and Medicine (Graduate), Operating Systems (Graduate), Machine Learning, Reinforcement Learning, Analysis of Algorithms

Harvard University

Summer 2017: Advanced Mobile Application Development Using Swift and iOS

Certifications & Credentials

MongoDB Certified Developer (C100DBA)	See Credential	2021
MongoDB Certified Developer (C100DEV)	See Credential	2021
Certified Kubernetes Administrator (CKA)	See Credential	2020
Certified Kubernetes Application Developer (CKAD)	See Credential	2019
Udacity Full Stack Developer Nanodegree	See Credential	2016
Udacity Front-End Web Developer Nanodegree	See Credential	2016
Udacity iOS Developer Nanodegree	See Credential	2016

Research Experience

University of Massachusetts Lowell

Rehabilitation for Outpatient Stroke (2019 – Present)

- Developed an algorithm to determine effectiveness of physical therapy for stroke patients
- Research paper in progress

Synthesizing Crowdsourced Physician Reviews (2020 – Present)

- Created a dashboard to consolidate physician ratings to aid patients in choosing a doctor

Technical Skills

Orchestration	Kubernetes, Helm, Docker, Docker Compose, Nginx
Cloud Providers	Microsoft Azure, Amazon Web Services, Google Cloud Platform
Languages	Node.JS, C#, Python, C++, Go, TypeScript, JavaScript, Swift, Java, C, MATLAB
Data Analysis	TensorFlow/Keras, PyTorch, Jupyter, MongoDB Aggregation, D3.js, Tableau
Databases	MongoDB, Redis, MySQL, Google Firebase
CI/CD	Git, GitHub, Azure DevOps, Atlassian Bitbucket

Awards & Honors

David E. Shulman Computer Science ACM Leadership Award	2020
Presenter – FIRST Robotics World Championship Conference in Detroit	2018
FIRST Robotics Dean's List Award Semi-Finalist	2017